

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\agent-a787af598b70d19b8.jsonl`  
- SHA-256: `b92f98426d77b6c7d97780234de4cef3996f2117f6e9a46554738ecfd5152020`  
- Source modified: `2026-07-03T16:42:35+00:00`  
- Imported at: `2026-07-05T16:48:26+00:00`  
- project: `subagents`  
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`
```

Transcript

1. user (2026-07-03T16:40:31.856Z)

You are a code-review finder agent. Review target: PR #451 of the badminton-highlight-indexer repo, checked out at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer on branch codex/alpha-readiness-plan (HEAD cd07339). Get the diff with: git diff origin/master..HEAD. Docs-only PR: new docs/ALPHA_LAUNCH_READINESS.md plus one-line index additions to docs/README.md, docs/DOC_STATUS.md, docs/NEXT_STEPS.md.

ANGLE ? conventions (CLAUDE.md). Find and read the CLAUDE.md files governing the changed files: C:/Users/avidu/.claude/CLAUDE.md (user-level), the repo root CLAUDE.md and CLAUDE.local.md at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/, and any docs/CLAUDE.md. Also check AGENTS.md at repo root if present. Check the diff for CLEAR violations of rules those files state ? e.g., doc-formatting rules, honesty/attribution rules, tracked-doc conventions, memory-linking conventions ([[...]] links). Only flag a violation when you can quote the exact rule and the exact line breaking it. No style preferences, no "spirit of the doc" inferences. In each finding name the CLAUDE.md path and quote the rule. If no CLAUDE.md applies or no rule is broken, return [].

Return up to 6 candidate findings as a JSON array of {file, line, summary, failure_scenario}. Your final message must be only the JSON array.

2. assistant (2026-07-03T16:40:33.811Z)

I'll help you review PR #451 for violations of conventions defined in CLAUDE.md files. Let me start by understanding the structure and reading the relevant documentation.

3. user (2026-07-03T16:40:34.564Z)

```
diff --git a/docs/ALPHA_LAUNCH_READINESS.md b/docs/ALPHA_LAUNCH_READINESS.md  
new file mode 100644  
index 0000000..4b3d49d  
--- /dev/null  
+++ b/docs/ALPHA_LAUNCH_READINESS.md  
@@ -0,0 +1,171 @@  
+# Alpha Launch Readiness Plan  
+  
+> **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:** 2026-07-03; **Last updated:** 2026-07-03  
+> **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`  
+> **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed in `docs/README.md` and `docs/DOC_STATUS.md`.  
+> **Relation to existing docs:** consolidates the launch-relevant pending items from `NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`, `COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.  
+> **Honesty note:** `[verified]` means checked against the local synced repos on 2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not legal, financial, or launch approval.
```

```

+
+---
+
+## 0. TL;DR
+
+The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but
the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has 15
badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven
commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]`
the n=15 quality baseline is stale and must be intentionally refreshed before it can gate
anything.
+
+The first controlled alpha should be a hosted, authenticated product loop: upload a video,
process it, select rallies, compile, and download a reel. The owned-model/commercial rally-
detection claim remains a separate NO-GO until the ship-gate target is ratified and the WASB
checkpoint provenance is resolved.
+
+## 1. Problem & goal
+
+The previous launch blockers were spread across roadmap docs and mixed together three different
questions:
+
+- **Can an invited tester use the product without SSH?**
+- **Can the 15-video golden corpus now support honest quality gates?**
+- **Can we claim or ship an owned/commercial rally detector?**
+
+This doc separates those into executable workstreams. The goal is to reach an invite-only alpha
where the product is usable end-to-end, the quality claims are honest and reproducible, and the
remaining model/legal gates are explicit rather than accidentally implied by launch activity.
+
+## 2. Decisions locked
+
+| # | Decision | Source / date | Implication |
+|---|---|---|---|
+| D1 | **Product alpha and owned-model launch are separate decisions.** | `NEXT_STEPS.md` NO-GO
banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled hosted product loop without
claiming the owned model is commercially cleared. |
+| D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
+| D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature
completeness, baseline refresh, and gate ratification come before quality claims. |
+| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-
windowing safety; keep it as an opt-in/analysis lever. |
+| D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.**
| sibling `khehsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha can use
manual/offline corpus promotion; automated flywheel follows after serving and gates are stable.
|
+
+## 3. Verified snapshot, 2026-07-03
+
+All five Git checkouts were fast-forwarded before analysis:
+
+| Repo | Branch | Synced HEAD |
+|---|---|---|
+| `badminton-highlight-indexer` | `master` | `e4914d1` |
+| `khehsutra` | `master` | `b0f70a5` |
+| `rally-annotator` | `main` | `ef9e088` |
+| `sports-data-collector` | `master` | `bfc014f` |
+| `sports-obsreport` | `main` | `a0c7252` |
+
+Corpus/eval facts:
+

```

```

+- `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says `output/human_lovo_manifest.json` is
the ingested golden corpus at `n = 15`.
+- `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest found
all 15 trajectories and labels in this workspace.
+- `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
+- `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only 6
videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
+- `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
+- `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
+- `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15> --features-dir
output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off tolF1 `0.1382`,
but marked the frozen baseline stale due to config/corpus changes.
+- `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity group`
currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
+
+### 4. Launch strategy
+
+### 4.1 Alpha lane: make the product usable
+
+The alpha lane is successful when an invited tester can open the app, authenticate, upload a
video, watch the job progress, pick rallies, compile a reel, and download it. This lane should
avoid model-quality promises beyond the measured state and should default to the most reliable
serving path.
+
+### 4.2 Quality lane: make the 15-video corpus gate-ready
+
+The quality lane is successful when the 15-video corpus is portable, reproducible, feature-
complete, and backed by refreshed baselines. Only then should R2 tolF1, Gen-0/TemporalMaxer, or
other model/promotion decisions be ratified.
+
+### 4.3 Flywheel lane: make tester feedback useful
+
+The flywheel lane is successful when user corrections can flow into the collector/vault path
with consent/provenance and a visible re-eval delta. This is valuable for alpha learning, but it
should not block the first controlled alpha if manual promotion can bridge the gap.
+
+### 5. Risks and guardrails
+
+| Risk | Why it matters | Guardrail |
+|---|---|---|
+| Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
+| Stale local manifests create false confidence | Manifest-driven tools skipped or failed until
paths were rebased | Make the corpus manifest portable and add a path/fixture smoke check |
+| 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still n=6
today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
+| Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly reported
stale baselines | Refresh baselines only after reviewer sign-off and record the command/output |
+| P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but can
consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha lane is
already stable |
+| WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained model
regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product launch
tasks |
+
+### 6. Honest limits
+
+Completing this doc's alpha lane does not prove the owned model is ready, that WASB weights

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are commercially clean, that the product can serve unbounded public traffic, that multi-sport is ready, or that every tester correction automatically improves the model.

+
+Completing the quality lane does ****not**** by itself deploy a service. It only makes the measurement surface honest enough for a launch decision.

+
+## 7. Deliverables & progress tracker

+
+Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the row explicitly names a cross-repo batch.

+
+| ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
+|----|-----|-----|-----|-----|-----|----|
+| A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-indexer` | - | No | IN PROGRESS | #451 |
+| A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified | `khesutra` + `badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
+| A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 | No | TODO | - |
+| A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` + vault | A2 | Owner upload/storage scope | TODO | - |
+| A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2 | Reviewer sign-off | TODO | - |
+| A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic\_lovo\_n6` floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |
+| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
+| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data availability | TODO | - |
+| A8 | Re-run `fusion\_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips | `badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
+| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner approval for M0.4 scope | TODO | - |
+| A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-indexer` | A4, A5, A8 | Owner decision | GATED | - |
+| A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred, document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance | GATED | - |
+| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train|reeval` job, and delta-F1 surface | `khesutra` + `badminton-highlight-indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |
+| A13 | Alpha launch review: runbook, privacy/consent copy, known limits, rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch decision | TODO | - |

+
+## 8. Concrete next PR sequence

+
+1. ****Docs PR:**** land this tracker.
+2. ****Corpus portability PR:**** rebase/portable manifest tooling plus missing-file smoke test.
+3. ****Hosted alpha PR/batch:**** health/auth/upload/process/job/compile/download verification and runbook updates.
+4. ****Baseline PR:**** intentional n=15 nightly baseline refresh and heuristic floor refresh.
+5. ****Ablation/fusion PRs:**** fix ablation import, complete 15-video features, rerun promotion/rejection decisions.
+6. ****Launch decision PR/docs update:**** record ship-gate target, C3 status, alpha copy limits, and launch checklist.

```

+## 9. Open questions
+
+1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or should
P10/P11 be reopened before any tester wave?
+2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU, GCE L4
VM, or another approved GPU host?
+3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains unresolved?
+4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
+5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every clip,
or should the eval loaders support mixed schemas with explicit capability tags?
+
+## 10. Definition of done
+
+- [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.
+- [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
+- [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely on stale
local absolute paths.
+- [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
+- [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
+- [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected on the
15-video corpus.
+- [ ] Ship-gate target and allowed alpha claims are recorded.
+- [ ] WASB C3 status is recorded and reflected in alpha messaging.
+- [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for alpha-
plus.
+- [ ] Owner records GO/NO-GO for the first alpha wave.
+
+## 11. References
+
+Internal docs:
+
+- `docs/NEXT_STEPS.md`
+- `docs/ROADMAP.md`
+- `docs/data_pipeline/GOLDEN_VIDEOS.md`
+- `docs/R2_EVAL_METRIC_DESIGN.md`
+- `docs/RALLY_DETECTION_FIXES_PLAN.md`
+- `docs/COMPUTE_DECOUPLED_SERVING/README.md`
+- sibling `khehsutra/NEXT_STEPS.md`
+- sibling `khehsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
+- sibling `khehsutra/docs/PER_RALLY_SELECTION.md`
+- sibling `sports-data-collector/data/golden_source_videos.json`
+
+Local verification commands used for the 2026-07-03 snapshot:
+
+```powershell
+python -m backend.eval.fusion_compare --features-dir output
+python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
+python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor
eval_baselines\heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
+python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir output
--no-report
+python -m backend.eval.ablation --features-dir output --granularity group
+```
+
+---
+
+#### Changelog
+
+- 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`, golden-corpus
docs, local eval artifacts, and the current 15-video corpus state.
diff --git a/docs/DOC_STATUS.md b/docs/DOC_STATUS.md
index 34411fd..2541672 100644
--- a/docs/DOC_STATUS.md

```

```

+++ b/docs/DOC_STATUS.md
@@ -30,6 +30,7 @@ Before a foundational/tracked doc moves to `docs/archives/`:
  ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's $7 is the completeness
  source-of-truth)
  | Doc | Status | Archive-ready? | Next action |
  |---|---|---|---|
+| `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created
  2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-
  model/commercial rally-detection gates | No | Execute $7 in order: corpus portability + n=15
  baselines, hosted alpha serving, ablation/fusion reruns, then launch G0/N0-G0 |
  | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M9 LARGELY SHIPPED**
  (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights
  pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual:
  counsel/prices/OAuth/real-run within the D17 cap) |
  | `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** ($7) | No | Drives the
  R-series; complete when $7.1 single quality number is attributed |
  | `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation
  pending) | Promote-or-reject toF1 as the gate (ablation) ? then archive with the R-cluster |
  diff --git a/docs/NEXT_STEPS.md b/docs/NEXT_STEPS.md
  index f186ff9..a9f35a8 100644
  --- a/docs/NEXT_STEPS.md
  +++ b/docs/NEXT_STEPS.md
  @@ -9,7 +9,7 @@

```

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  ## ? Prioritized pending ? all tracks
  **P0 ? highest leverage / unblocks the most**
-0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
  lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only**
  (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/`
  SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service**
  (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA
  ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll
  `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-
  OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our
  cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:
  [[ui-driven-cloud-serving-gap]].
+0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
  lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only**
  (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/`
  SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service**
  (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA
  ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll
  `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-
  OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our
  cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:
  [[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:**
  `docs/ALPHA_LAUNCH_READINESS.md` $7.
  0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-
  included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install`
  ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state
  / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b,
  P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** ***(1)** P3 **GPU validation +
  screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB
  0-rally tuned-config blocker), ***(2)** P2b-install (uv/script + truly-local default config),
  ***(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud
  (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-
  session-packaging-app]]`.
  1. ? **`COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified).
  M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-
  OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All $8 gating Qs
  locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA
  (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF
  team-domain/AUD + ingress-lock; real cloud verification authorized within a **$100/?10k cap**
  (D17).
  2. **[owner, ongoing] Grow the golden corpus ? multi-court badminton + ?3 matches/sport (TT

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first). The single highest-leverage move; the quality track *and* full-corpus training feed on
it.
diff --git a/docs/README.md b/docs/README.md
index 0df1a69..58be5ff 100644
--- a/docs/README.md
+++ b/docs/README.md
@@ -15,6 +15,7 @@
- [DOC_STATUS.md](DOC_STATUS.md) ? **? living Documentation Status & Archival Register**: every
foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence
checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here;
update rows as docs complete.

## Product / platform plans
+- [ALPHA_LAUNCH_READINESS.md](ALPHA_LAUNCH_READINESS.md) ? **? Active tracked project (`DRAFT`,
created 2026-07-03):** cross-repo alpha readiness tracker after the golden corpus reached 15
videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel) from the
owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved). $7 is the
source of truth for the concrete next PR sequence: corpus portability, n=15 baseline refresh,
ablation/fusion reruns, hosted serving, alpha copy/launch review.
- [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked
project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to
DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never
branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend`
(what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU,
in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run
GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI).
Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY
proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a
[runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage
ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). $7 M0?M10 tracker is the source
of truth.
- [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED +
ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing
bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB
runner**, **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ?
DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving
endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving"
subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under
`docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).
- [PACKAGING_PLAN.md](PACKAGING_PLAN.md) ? **? Active tracked project (`IN PROGRESS`, 11 PRs
merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ?
`pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA**
single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision
D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP_APP_PLAN.md`
narrative.** $7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **?
Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4
(ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-
app]]`.

```

4. assistant (2026-07-03T16:40:36.654Z)

Now let me read the CLAUDE.md files to understand the conventions I need to check against.

5. user (2026-07-03T16:40:37.102Z)

```

1      # Global instructions (all projects)
2
3      ## Repo-freshness convention: git pull BEFORE reading
4      **Always `git pull` a repo before starting to read it** ? at session start for the
primary
5      working directory, and again for any sibling/local repo the moment work touches it.
6
7      - **Why:** the owner closes every session on a "clean slate", but multiple parallel
slates
8      (sessions/machines) exist ? PRs get merged and master moves between sessions, so the

```

```

local
9      checkout can silently lag the remote truth. Pull first so ramp-up reads (handoff,
docs, code)
10     reflect where the remote actually is.
11     - **How (safe form):** `git pull --ff-only` on the current branch (a plain fetch+ff).
Never
12     auto-merge, rebase, or discard local state to make a pull succeed.
13     - **If it can't fast-forward** (diverged branch, dirty tree in the way): do NOT force
anything ?
14     report the state to the owner and proceed read-only on what's local, flagging that it
may be
15     stale. Uncommitted changes are often a sibling session's or the owner's WIP; never
clobber them.
16
17     ## Git branching safety: concurrent sessions / shared checkouts
18     Multiple sessions ? and spawned/background tasks ? may share ONE working checkout and
create
19     branches + commit **concurrently**, so `git branch` / `git log` can change UNDER you
between two
20     commands. (Spawned "fresh worktree" tasks are NOT always isolated.) This has caused a
real
21     incident: a sibling commit landed in the gap between a branch-point check and `git
checkout -b`,
22     so the new branch rode on top of it and a **squash-merge silently absorbed the sibling's
work**.
23     Protocol ? do this EVERY time, not only when you suspect a collision:
24     - **Branch from the REMOTE, explicitly:** `git fetch origin` then `git checkout -b
<name> origin/<base>`
25     (e.g. `origin/master`). Never assume the current branch is the intended base.
26     - **Re-verify right before `git add`/commit:** `git branch --show-current` + `git log
--oneline -1`
27     (HEAD is yours). Before opening a PR, `git log --oneline origin/<base>..HEAD` must
list ONLY your
28     commits ? if a sibling commit appears, re-cut the branch from `origin/<base>`.
29     - **Stage explicit paths** (`git add <files>`) ? never `git add -A`/`.`/`-u`, so sibling
or untracked
30     files can't ride into your commit.
31     - **Serialize repo writes:** don't start a repo-writing spawned/background task while
you hold
32     uncommitted work ? commit or push first. If one may be running, treat the tree +
current branch
33     as able to shift, and put this branching-safety reminder into the spawned task's own
prompt.
34
35     ## Session-handoff convention
36     Maintain a living **session handoff** for each project and use it to resume context
quickly
37     across windows.
38
39     - **Where it lives:**
40     - If the project has an auto-memory dir (`~/ .claude/projects/<project>/memory/`), keep
the
41     handoff as `session-handoff.md` there, and list it under a ***"? Start here"*** entry
at the
42     top of that project's `MEMORY.md`.
43     - Otherwise, keep a `SESSION_HANDOFF.md` at the repo root.
44     - **What it contains** (keep it to ~1 page ? it POINTS to detailed artifacts, never
duplicates them):
45     1. **You are here** ? current state.
46     2. **Next steps / open threads.**
47     3. **Ramp-up kit** ? the exact files/docs to read to get current.
48     4. **Key decisions** ? so they aren't relitigated.
49     - **At session start:** if a handoff exists, read it before starting substantive work,
and use it
50     (plus its ramp-up kit) to get up to speed instead of replaying past conversation.

```



```

51      - **Updating it:** keep it current **automatically** ? refresh at meaningful checkpoints
(a PR is
52      pushed/merged, a key decision is made, work shifts phase, or a session is wrapping
up), so a
53      restart can ramp up without losing context. Do **not** rewrite it every turn ? only
when
54      something worth resuming from changed. The phrases **"update the handoff"** /
**"refresh the
55      handoff"** force an immediate full rewrite on demand.
56      - **Keep it honest:** it reflects real, shipped state ? never aspirational. (See each
project's
57      attribution/working-agreement note where present.)
58
59      ## Code review ? pre-LGTM gate (applies to every PR/code review, every project)
60
61      Do **not** post **LGTM** or otherwise sign off on a pull request until I have personally
**verified** the following ? never trust the PR description's claims, reproduce them locally:
62
63      1. **Full test suite is green.** Run the whole suite (e.g. `run_all_tests.py` /
`pytest`), not just the changed files.
64      2. **Install any dependency needed** so the suite actually collects and runs ? do not
let modules skip or error for missing deps. (If a check genuinely can't run locally ?
GPU/WSL/real-video/etc. ? say so explicitly and note what was and wasn't verified.)
65      3. **Lint + type checks are clean** ? `ruff check .` and `mypy ...`, matching the repo's
CI. Install `ruff`/`mypy` if absent.
66      4. **Coverage stays up** ? run with coverage and confirm it stays **at or above the
repo's floor** (e.g. the CI `--cov-fail-under` value) and does **not regress** versus the base
branch.
67      5. **Review findings are resolved** ? bugs and repo-convention/discipline issues
addressed.
68
69      LGTM is a *verified* stamp, never a rubber stamp. State plainly what was and wasn't
verifiable locally.
70

```

6. user (2026-07-03T16:40:37.730Z)

```

1      # CLAUDE.md ? project memory / agent handoff
2
3      Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:
4      FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
5      repos: `sports-data-collector`, a VLC-based `rally-annotator`.
6
7      ## Read these first
8      - **docs/README.md** ? doc index. **docs/CODE_MAP.md** ? code navigation + data
9      contracts (the cold-start map; read before touching code).
10     - **Authoritative owned-model roadmap:** `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+
the
11     live `docs/NEXT_STEPS.md`). These SUPERSEDE the 2026-06-07 strategy docs below where
they
12     disagree (licensing, Gemma-4 stance, base model, conversational-QA).
13     - **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`,
`docs/PMF_MARKET_RESEARCH.md`,
14     `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also
`docs/ROADMAP.md`
15     (offline-segmenter narrative) and `docs/BACKLOG.md`.
16     - **Rally-detection quality track (design merged 2026-06-13, owner-gated, implementation
in progress):**
17     `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model) ?
`docs/MULTI_SIGNAL_FUSION_PLAN.md`
18     (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built
`rally_gate.py`
19     "Gate A" ? implemented in `backend/eval/served_gate_a.py`, default-OFF) ?
`docs/EXPERIMENT_HARNESS.md` (A/B + shadow
20     eval so cues land flag-gated/default-OFF with zero regression) ?

```

```

`docs/ROORA_LABELING_GUIDELINES.md`
21     (v8 vendor labeling spec). **`docs/NEXT_STEPS.md` has the prioritized pick-up for this
track.**
22     - **History:** `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and
23     **`checkpoints/`** (latest: `2026-06-strategy-foundation/`). **Archive policy:** only
move
24     **terminal-state** (DONE/REJECTED) work; live docs stay at `docs/` top level ? and
**before
25     archiving, HONESTLY update the doc first** (verify claims vs code ? flip status + a
completion
26     note ? update inbound refs ? keep the lineage ? then `git mv`), per the checklist +
living
27     lifecycle register in **`docs/DOC_STATUS.md`**.
28
29     ## Current state (July 2026)
30     - **Platform slice SHIPPED:** async job model (`backend/api/job_worker.py`, #339),
31     `StorageBackend` + `ComputeBackend` registries with concrete implementations
32     (`backend/storage/` ? local/gcs/gdrive/gdrive_api/s3; `backend/compute/` ?
33     local/cloud_run), capabilities/assets endpoints, and the data flywheel
34     (`backend/flywheel.py`). `COMPUTE_DECOUPLED_SERVING` M0?M9 largely shipped
35     (ADR-014 ratified 2026-06-21); Gen-0 owned-model weights pushed (`weights/0.1.0-l4/`).
36     - **Launch decision: NO-GO (owner, 2026-06-30, #404)** ? the rally-detection product is
37     NOT cleared for external/commercial launch; we stay in development. Two unblockers
gate a
38     future GO: (1) set + meet the ship-gate target (#172, needs corpus growth); (2)
resolve
39     WASB shuttle-detector weight provenance (commercial-clean licensing).
40     - **Active tracks** (see `docs/NEXT_STEPS.md` for the prioritized pending list):
41     rally-quality / owned-model ML work (gated on golden-corpus growth), packaging
42     (`docs/PACKAGING_PLAN.md`), and the khelsutra SPA/deploy sibling track.
43
44     ## Architecture in one breath
45     - **Pluggable registries** (ABC + `Registry` singleton): segmenters / providers /
validators
46     / sports / annotations ? see `backend/pipeline/segmenters/base.py`.
47     `StorageBackend` + `ComputeBackend` registries are **implemented**
(`backend/storage/`,
48     `backend/compute/`; PLATFORM S3 ? realized by `COMPUTE_DECOUPLED_SERVING`).
49     - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
50     - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
51     user_version` migration ladder in `backend/infrastructure/database.py`.
52
53     ## Committed guardrails (non-negotiable)
54     - **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
55     training data source** (terms/copyright). Commercial-clean licensing for EVERY
dependency ?
56     code, data, AND weights: **MIT / Apache-2.0 / BSD only** (ratified 2026-06-09, owner-
agreed).
57     - **Owned-model base:** **Qwen3-VL (Apache-2.0)** primary, InternVL3 (MIT) fallback;
**Gemma-4
58     is AMBER / bench-only** (Apache grant likely but Prohibited-Use posture unconfirmed ?
counsel
59     needed; do NOT build on it without sign-off). **Reject Gemma 3** (viral license).
**Exclude
60     minors** from the training pool; per-user training data needs a separate explicit opt-
in.
61     - **v1 = rally + highlight + history + correction loop**; **gait/biometrics strictly
future**
62     (GDPR Art. 9).
63     - **Generic data schema**; `coach ? athlete` is an **optional overlay**, not baked in.
64     - **Compute = GCP ONLY (ADR-013, 2026-06-20):** GPU training + serving run on **GCP**
(`deploy/gcp/`;
65     Cloud Run+GPU for serving per ADR-012). **DigitalOcean is DROPPED for compute** (DO
GPU not enabled

```

```

66         on the account + not cleanly credit-funded; Paperspace is subscription-gated). Don't
re-evaluate
67         DO/Paperspace for compute without a new ADR; the `$200` DO credits go to **non-compute
only**. The
68         `deploy/digitalocean/` scripts are kept solely as provider-agnostic Linux-GPU setup
reused by GCP.
69
70     ## Workflow conventions
71     - **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
72     - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.10+ (the `trackers`
package requires ?3.10); `ffmpeg` required.
73     - Never push to the default branch directly; descriptive commit messages.
74     - **Substantial design/project work gets a tracked project doc** following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` §7,
`docs/GOLDEN_REGRESSION_FIXTURES.md`.
75     - **Launch / "bit-flip" PRs** (a default-flip that drives a tracked project to terminal
state) MUST, in the SAME PR, run the `docs/DOC_STATUS.md` §2 archival due-diligence on that
project's docs (verify-vs-code ? flip status + completion note ? update inbound refs ? keep ADR
lineage ? `git mv` to `docs/archives/` ? record in DOC_STATUS §3), and fix any internal-link
breakage the move causes. Enforced by `.github/PULL_REQUEST_TEMPLATE.md` + the `link-check` CI
job (`python -m tools.check_md_links`).
76     - ?? Hand the heaviest machine-side checks (real video, the full GPU/WSL/WASB suite) to
the owner,
77         who has the golden corpus + labelled videos; verify pure-logic paths locally.
78

```

7. user (2026-07-03T16:40:39.071Z)

(Bash completed with no output)

8. assistant (2026-07-03T16:40:41.000Z)

Let me check for additional CLAUDE files and review the specific changes in detail.

9. user (2026-07-03T16:40:42.613Z)

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/CLAUDE.md

10. assistant (2026-07-03T16:40:44.461Z)

Now let me check the specific files mentioned in the diff and verify the indexing and conventions.

11. user (2026-07-03T16:40:44.921Z)

```

1     - [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) ? *** Foundation doc** for
closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an
**adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-
sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with
falsifiable success criteria**, and (§7) a **tracked project plan** ? work-item checklist,
sequencing, the single quality **number** attributed to the effort, and a **definition of done**
(project completes when all tiers land + the number is recorded). The plan that the
[QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) log executes against; ties together
EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
OWNED_MODEL_IMPLEMENTATION_PLAN.
2     - [RALLY_DETECTION_GAP_CLOSURE.md](RALLY_DETECTION_GAP_CLOSURE.md) ? **Tracked execution
doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before
launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it:
boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini

```

both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally\_gate` Gate A wiring, complementary fusion. **Peer** of `RALLY\_QUALITY\_RESEARCH.md` (execution tracker, not research); §7 is the source of truth.

3 - PROMINENT_COURT_DETECTION.md ? **Design + experiment (2026-06-19, default-OFF, owner-gated flip)** for the **multicourt G2 over-fire**: identify the **prominent (foreground) court** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` *before* windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? **pseudo-3D play-volume** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion\_features.point\_in\_polygon` + `FusionConfig.court\_polygon`. §8 progress tracker is the source of truth.

4 - RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.09170.323 at n=11, p=0.002; merge-rate 0.69470.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY\_QUALITY\_RESEARCH.md` + `QUALITY\_ITERATIONS.md`.

5 - EVALUATION.md ? How we measure and improve quality.

6 - EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.

7 - QUALITY_ITERATIONS.md ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.

8 - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT\_HARNESS.md` and `MULTI\_SIGNAL\_FUSION\_PLAN.md` (ADR-007 modular rally layer).

9 - PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization feature on a generalization-first baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *a tool that gets better the more you help it*). Records the **locked owner decisions** (identity, storage, `min\_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

10
11 - GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started)**: video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends [GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md).

12

13 ## ? Documentation status & governance

14 - [AUDIT_REPORT_khelsutra-guru_2026-07-02 (? COMPLETE / ARCHIVED)](archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md) ? the five-repo `khelsutra-guru` workspace code audit + its folded-in **Remediation tracker** (P0?P28 all closed). **Archived 2026-07-02** once every remediation wave landed and the last residuals were closed or filed as issues (#443 CORS, #444 corpus Phase-2, #445 GCS eval; + #301/#327/#332/#384). Absorbed the prior

`CODE\_CLEANUP\_AUDIT\_2026-06-24` + June-2026 hardening sweep (both archived).

15 - DOC_STATUS.md ? *** living Documentation Status & Archival Register***: every foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here; update rows as docs complete.

16

17 ## Product / platform plans

18 - ALPHA_LAUNCH_READINESS.md ? *** Active tracked project (`DRAFT`, created 2026-07-03):** cross-repo alpha readiness tracker after the golden corpus reached 15 videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel) from the owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved). \$7 is the source of truth for the concrete next PR sequence: corpus portability, n=15 baseline refresh, ablation/fusion reruns, hosted serving, alpha copy/launch review.

19 - [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? *** Active tracked project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend` (what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU, in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for \$0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). \$7 M0?M10 tracker is the source of truth.

20 - [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED + ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past\_projects/DECOUPLED\_COMPUTE/` (incl. `DEPLOY\_REPORT\_GCP\_2026-06-20.md`).

21 - PACKAGING_PLAN.md ? *** Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA** single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP\_APP\_PLAN.md` narrative.** \$7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). *** Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

22 - DESKTOP_APP_PLAN.md ? *** Superseded narrative (by `PACKAGING\_PLAN.md`; D10 = self-hosted webserver, NOT a desktop app).** Original scoping doc (2026-06-18, `[design]`): package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4x real-time on a laptop 2070S)? \$7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

23

24 ## Practical / ongoing

25 - CREATING_REELS.md ? *** End-user guide:** install the LIGHT-tier app ? `indexer --app` ? process a video ? pick rallies ? download a watermarked reel. Task-first usage walkthrough (web UI + the API chain), with a \$6 pointer to the cloud compute option. LIGHT tier only (offline `motion` / optional Gemini; no on-device GPU or one-click cloud). Install mechanics ? root `[README.md](../README.md); packaging roadmap ? PACKAGING_PLAN.md.

26 - CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md ? ***

Operator runbook:** `gs://` bucket videos ? highlight reels on the approved **L4 Cloud Run** (scale-to-zero, \$0 idle). Copy-pasteable: build image ? create the GPU Job ? cloud-serving control plane ? process(cloud)?query?compile?download ? prove the path (``verify_compute_path.py``) ? \$0 teardown. **Verified e2e 2026-06-23.** Raw deploy knobs ? ``../deploy/cloudrun/README.md``.

27 - ``SETUP_NEW_MACHINE.md`` ? **? The canonical "spin up a box ? ready for training & inference" runbook. ** Copy-pasteable, human-followable, **verified end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)** , see [``deploy/gcp/README.md``](``../deploy/gcp/README.md``): credentials ? provision ? bootstrap ? suite (922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training, with pointers to the GPU/native-detector + bucket paths. The ``deploy/digitalocean/*`` scripts are provider-agnostic and reused on GCP.

28 - ``DATA_IN_GCS.md`` ? **? The corpus source-of-truth map: where every piece of data lives in `gs://khelsutra-rally-corpus`** (canonical layout · current contents + drift · how ``backend.worker`` consumes it · the still-local-only **gap** · migration plan). Read this so a session pulls data from the bucket, not from someone's ``C:/F:/Drive`` ? the goal is to remove the local box as a blocker. Pairs with the two runbooks below.

29 - [``deploy/gcp/nightly_regression/README.md``](``../deploy/gcp/nightly_regression/README.md``) ? **? Nightly engine-regression OPERATIONS MANUAL (one-stop shop):** enable · run on demand · change any setting (email recipients/frequency/clips/config) · observe · pause/disable/remove. The full WASB-pipeline reel-count+cost+email nightly on an ephemeral GCP GPU VM (design/status: ``NIGHTLY_ENGINE_REGRESSION.md``). The lighter CPU cached-trajectory tripwire is ``scripts/nightly_regression.py`` (local Windows Scheduled Task, PR #202).

30 - ``TRACKNET_WSL_SETUP.md``

31 - Evaluation harness and calibration drivers (in ``backend/eval/``)

32

33 ## Data layer (collection · annotation · golden sets ? **no ML**)

34 - [``data_pipeline/``](``data_pipeline/README.md``) ? **the isolated data pillar** : how footage is recorded (``VIDEO_RECORDING_GUIDELINES``), golden sets are created + which videos (``GOLDEN_SET_IMPLEMENTATION`` · ``GOLDEN_SET_PHASE1_VIDEOS`` · ``GOLDEN_VIDEOS``), and data is shared/ingested (``GOLDEN_DATA_SHARING``). Deliberately **no segmentation/stitching/model** detail ? that's the ML pillar (top level + ``COMPUTE_DECOUPLED_SERVING/``). The **annotation-vendor (Roora)** engagement is archived under ``archives/roora-vendor/``.

35

36 ## Conventions & templates

37 - ``PROJECT_DOC_TEMPLATE.md`` ? Reusable skeleton for a **tracked project doc** : status header · decisions-locked · progress tracker (??/?) · definition-of-done · archived on completion. Standardizes the emergent pattern in ``RALLY_QUALITY_RESEARCH.md`` §7 and ``GOLDEN_REGRESSION_FIXTURES.md``. Use for any substantial design/project work.

38

39 ## Historical documents

40 Completed plans, old research, ADRs, and checkpoints live in ``archives/``.

41

42 - **June-2026 test-hardening sweep** ? **COMPLETED** (2026-06-14; T1?T5 + full loop + verification per #162) and **archived** ; the former top-level pointer-stubs were removed 2026-07-02 (absorbed into the umbrella's *Prior audits*). Full records: ``archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md``, ``archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md``, ``archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md``.

43 - **CODE_CLEANUP_AUDIT (2026-06-24)** ? the prior deep engine audit (35 findings; Phase 1 complete, Phase 2 largely shipped, remaining = the ``main.py`` god-module split). **Absorbed into the umbrella** (see its *Prior audits (absorbed)*) and archived ? ``archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md``.

44 - **KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN** ? the five-repo audit execution board, **folded into the umbrella** as its *Remediation tracker (live)* and archived ? ``archives/past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md``.

45 - ``DECISION_SNAPSHOT_REGRESSION.md`` ? **? Tracked project (decisions D1?D11 LOCKED 2026-06-25, #383; implementation P1?P7 pending):** a deterministic **quality-neutral proof** ? freeze + compare the pipeline's **decision output**

(rally segments + stitch/concat plan) off a cached trajectory, GPU-free, in CI. The behavioral proof that lets a *genuine* (non-AST-identical) refactor be confidently called quality-neutral. **Never hashes the encoded reel** (non-deterministic). Extends the existing characterization snapshot (`test\_golden\_fixtures.py`) from segmentation to the stitch plan; peer of the M4 quality-floor + GPU `nightly\_reelcount.py`. §7 tracker is the source of truth.

46
47 - **Field-ops / customer-feedback** (recruiting flyer · field-visit playbook · PMF) ?
[`archives/field-pmf/`](archives/field-pmf/) (archived 2026-06-21; PMF validation still an active *business* thread ? see `NEXT\_STEPS.md`).
48 - **Roora annotation-vendor** engagement (labeling brief · vendor bake-off) ?
[`archives/roora-vendor/`](archives/roora-vendor/) (archived 2026-06-21).
49 - Superseded: per-video output layout ? [`archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`](archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md) (?
`PER\_VIDEO\_WORKSPACE.md`).
50
51 See `archives/README.md` for the umbrella policy and `archives/past\_projects/README.md`
for classification of small tech debt / starter projects.
52

12. user (2026-07-03T16:40:45.145Z)

1 # Documentation Status & Archival Register
2
3 > **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-diligence**. **Last full review:** 2026-07-02.
4 > **Update cadence:** whenever a doc changes lifecycle state (created / shipped a milestone / delivered / archived), and re-review at each clean-slate.
5 > **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update it as work finishes** to prove we're done, and (c) **never archive a foundational doc without honestly updating it first** (owner directive, 2026-06-21).
6
7 ---
8
9 ## 0. How to use this
10 - **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention. §3 is the full register.
11 - **Marking something complete?** Update the doc's own **§7 progress tracker** (the source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2 archival checklist.
12 - **This register does NOT duplicate completeness detail** ? it indexes lifecycle + archive-readiness and points to each tracked doc's §7.
13
14 ## 1. Lifecycle legend
15 `DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal record of a run/launch), **CONVENTION** (a standing rule/guide).
16
17 ## 2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git mv`)*
18 Before a foundational/tracked doc moves to `docs/archives/`:
19 1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts `[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it first.)
20 2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped, where (PRs, paths, artifacts), and what was **deferred / carried forward**.
21 3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`, `docs/NEXT\_STEPS.md`, and any cross-doc links ? the new path + status.
22 4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.
23 5. **`git mv` the whole dir/file ? `docs/archives/past\_projects/`** (terminal-state ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this in the doc's header).
24 6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.
25
26 > **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5

delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward WASB-C3/DPA/`md5`), reindexed `docs/README` + `past\_projects/README`, kept the ADR-010?014 lineage, then `git mv` to `docs/archives/past\_projects/DECOUPLED\_COMPUTE/`.

27

28 **## 3. The register**

29

30 **### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's \$7 is the completeness source-of-truth)**

31 | Doc | Status | Archive-ready? | Next action |

32 |---|---|---|---|

33 | `ALPHA\_LAUNCH\_READINESS.md` | ****DRAFT**** ? cross-repo alpha readiness tracker created 2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-model/commercial rally-detection gates | No | Execute \$7 in order: corpus portability + n=15 baselines, hosted alpha serving, ablation/fusion reruns, then launch G0/N0-G0 |

34 | `COMPUTE\_DECOUPLED\_SERVING/` | ****IN PROGRESS**** ? M0 approved + ****M1?M9 LARGELY SHIPPED**** (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap) |

35 | `RALLY\_QUALITY\_RESEARCH.md` | Foundation, ****tracked & ACTIVE**** (\$7) | No | Drives the R-series; complete when \$7.1 single quality number is attributed |

36 | `R2\_EVAL\_METRIC\_DESIGN.md` | **IMPLEMENTED** (augmenting) | No (gates R4 + dual-eval; ablation pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |

37 | `R1\_MILESTONE\_LAUNCH\_REPORT.md` | **LAUNCHED** (#204) ? terminal record | Soon (with the R-cluster) | Keep with R-series for narrative; archive when the track wraps |

38 | `RALLY\_DETECTION\_GAP\_CLOSURE.md` | **DRAFT** (owner-gated) | No | G1?G3 levers; part of the R/quality cluster |

39 | `RALLY\_DETECTION\_FIXES\_PLAN.md` | ****DONE**** ? findings A/B/C resolved; P2b promoted `inactivity\_temporal\_density` default-ON (#396) | With the R-cluster (deferred per its header; gate = R2 tolF1 ablation + \$7.1 number) | Archive with the R-series when the track wraps |

40 | `RALLY\_DETECTION\_CODE\_REVIEW.md` | Terminal snapshot ? findings A (#394), B (#394 ? promoted #396), C (`sdc#50`) all resolved | With the R-cluster | Archive with the R-series |

41 | `PER\_VIDEO\_WORKSPACE.md` | **DRAFT, IN PROGRESS** (P0 = #264) | No | ? supersede-reconcile (finding #1) **DONE**; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once) |

42 | `MULTI\_SIGNAL\_FUSION\_PLAN.md` | P1 **DONE**; P2+ owner-gated | No | P3/P4 (learned fusion, audio) ? needs GPU golden features |

43 | `EXPERIMENT\_HARNESS.md` | P1 **IMPLEMENTED** | No | A/B + promotion gate; supports the quality track |

44 | `PROMINENT\_COURT\_DETECTION.md` | **DRAFT** (owner-gated) | No | C-series; multi-track selection design |

45 | `EXECUTION\_POLICY.md` | P1 **SHIPPED**; P3b next | No | Async job policy; reused by Cloud Run dispatch |

46 | `GOLDEN\_REGRESSION\_FIXTURES.md` | **IN PROGRESS** (owner-gated) | No | Golden eval fixtures |

47 | ****Data layer**** ? `docs/data\_pipeline/` (`GOLDEN\_SET\_IMPLEMENTATION` . `GOLDEN\_SET\_PHASE1\_VIDEOS` . `GOLDEN\_VIDEOS` . `VIDEO\_RECORDING\_GUIDELINES` . `GOLDEN\_DATA\_SHARING`) | **LIVING** (data, ****no-ML****) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data\_pipeline/README.md`. Roora **vendor** docs archived ? `archives/roora-vendor/`. |

48 | `archives/past\_projects/AUDIT\_REPORT\_khelsutra-guru\_2026-07-02-COMPLETED-2026-07-02.md` ****umbrella**** | ****? COMPLETE / ARCHIVED (2026-07-02)**** ? the five-repo code audit + folded-in **Remediation tracker** (P0?P28 all closed) + absorbed prior audits. All non-owner-gated residuals landed; owner-gated/future items filed as issues (#443/#444/#445, + #301/#327/#332/#384). | Archived | Owner confirmed archival 2026-07-02; moved to `archives/past\_projects/` per \$2. Residuals now live as GitHub issues, not in this doc. |

49 | `DECISION\_SNAPSHOT\_REGRESSION.md` | ****DRAFT ? decisions LOCKED**** (D1?D11 locked 2026-06-25, #383). Zero code shipped for P1?P7. | No | Start P1+P2 (schema + stitch-plan factor); de-risk the P10 main.py split |

50 | `DESKTOP\_APP\_PLAN.md` | **DRAFT** (owner-gated, nothing built) | No | P0 compute-feasibility spike; relates to truly-local profile |

51 | `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` | **PLAN/roadmap** (owner-gated) | No (authoritative roadmap) | Gen-0?Gen-2; the owned-model track |

52 | `PERSONALIZATION\_PLAN.md` | **DESIGN** ? owner-adopted; Phase-0 | No | Per-user models (DB v5 store landed) |

53 | `UI\_ALPHA\_EXECUTION\_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |

54 | `I18N\_PLAN.md` | Design (P0); **in implementation** (adapted) | No | Executed in
`khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-
gated |

55 | `POST\_166...RISK\_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |

56

57 ### 3b. Foundational / strategy references (LIVING ? archive only if **superseded**, not
when "done")

58 | Doc | Status | Note |

59 |---|---|---|

60 | `PLATFORM\_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend`
registries ? realized by COMPUTE_DECOUPLED_SERVING |

61 | `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |

62 | `VIDEO\_LOCALITY\_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |

63 | `HOSTING\_PLAN.md` | PROPOSAL | **? partly overtaken by ADR-013 + the shipped site ?
verify (finding #5)** |

64 | `UI\_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half **in implementation** ?
`khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc)
|

65 | `STORAGE\_SHARING\_MODEL.md` | DRAFT (co-design pending) | ? |

66 | `ANALYZERS\_AND\_RECONCILERS.md` | DESIGN (owner-gated) | ? |

67 | `ROADMAP.md` / `CODE\_MAP.md` / `HOW\_RALLY\_DETECTION\_WORKS.md` /
`KHELSUTRA\_PRODUCT\_OVERVIEW.md` | LIVING references | Keep current; not archive targets |

68

69 ### 3c. Reports & terminal records (archive candidates once their track wraps)

70 | Doc | Status | Note |

71 |---|---|---|

72 | `REAL\_FOOTAGE\_VALIDATION\_REPORT.md` | validation complete | Motivates R2; archive with
the R-cluster |

73 | `RALLY\_DETECTION\_QUALITY\_REPORT.md` | report | Quality snapshot; part of R-cluster |

74 | `QUALITY\_ITERATIONS.md` | LIVING empirical log | **Keep** ? the paper-backbone
ablation log ([[paper-coauthor-aim]]) |

75 | `CODE\_AUDIT\_AND\_TEST\_HARDENING` . `HARDENING\_LOOP\_HANDOFF` . `DEFERRED\_HARDENING\_PLAN`
| COMPLETED ? **finish-archived 2026-07-02** (top-level stubs removed; absorbed into the
umbrella's *Prior audits*) | ? Full copies under `docs/archives/\*-COMPLETED-2026-06-14.md`;
inbound links repointed (finding #4 resolved) |

76

77 ### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

78 `PROJECT\_DOC\_TEMPLATE.md` . `NEXT\_STEPS.md` (? refreshed 2026-06-21) . `BACKLOG.md` .
`EVALUATION.md` . `ABLATION\_LAYER.md` . `SETUP\_NEW\_MACHINE.md` . **`CREATING\_REELS.md`** (? end-
user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) .
`CONVERT\_GCS\_VIDEOS\_TO\_HIGHLIGHTS.md` (? operator runbook ? `gs://` videos ? reels on L4
Cloud Run; verified e2e 2026-06-23) . `TRACKNET\_WSL\_SETUP.md` . `README.md` . *(2026-06-21
relocation: the data-layer guides moved ? `data\_pipeline/`; the field-ops + Roora docs ?
`archives/{field-pmf,roora-vendor}/`.)*

79

80 ### 3e. Already archived (`docs/archives/`)

81 `past\_projects/DECOUPLED\_COMPUTE/` (? 2026-06-21, #270) . `past_projects/low-regret-
moves/` . \*\*`field-pmf/`\*\* (field-ops + customer-feedback, 2026-06-21) . \*\*`roora-vendor/`**
(annotation-vendor engagement, 2026-06-21) . `PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`
 . `CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` . `DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md` . `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` .
`COMMERCIAL\_READINESS\_REVIEW.md` . `MONETIZATION\_AUDIT.md` . `TEST\_RUN\_SUMMARY.md` . `research/`
 . `checkpoints/` . `quality-iterations/` .
`past\_projects/CODE\_CLEANUP\_AUDIT\_2026-06-24-SUPERSEDED-2026-07-02.md` (absorbed ? umbrella) .
`past\_projects/KHELSUTRA\_GURU\_HEALTH\_REMEDIATION\_PLAN-FOLDED-2026-07-02.md` (folded ? umbrella).

82

83 ### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`

84 **ADR-001 ? ADR-014** (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ?
009 ? 010 ? 011 ? 012 ? 013 ? 014** . ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for
COMPUTE_DECOUPLED_SERVING).

85

86 ## 4. Open due-diligence findings / actions (2026-06-21 review)

87

88 > **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback

? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data layer** carved into `docs/data\_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data\_pipeline` one sets the "no segmentation/stitching" boundary).

89 1. **RESOLVED** (2026-06-21) ? `PER\_VIDEO\_WORKSPACE.md` supersedes
 `PER\_VIDEO\_OUTPUT\_LAYOUT.md`. **Folded forward** the `video\_slug` naming option (workspace §8.5) +
 the `output/GX010125\_run/` manual-prototype precedent; predecessor archived ?
 `docs/archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE\_MAP.md` repointed.

90 2. **RESOLVED** (2026-06-21) ? `NEXT\_STEPS.md` refreshed **with a current 2026-06-21**
 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is
 kept (clearly marked) as that track's detail.

91 3. **R-series / quality cluster is ACTIVE** (`R1`, `R2`, `RALLY\_DETECTION\_GAP\_CLOSURE`, `RALLY\_QUALITY\_RESEARCH`, `QUALITY\_ITERATIONS`, `REAL\_FOOTAGE\_VALIDATION\_REPORT`, `RALLY\_DETECTION\_QUALITY\_REPORT`). **Do NOT archive yet** ? archive as a **cluster** once R2 is promoted-or-rejected
 as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is",
 2026-06-21.)

92 4. **RESOLVED** (2026-07-02) ? finish-archived the hardening stubs. **The three top-**
 level pointer-stubs (`CODE\_AUDIT\_AND\_TEST\_HARDENING`, `HARDENING\_LOOP\_HANDOFF`,
 `DEFERRED\_HARDENING\_PLAN`) were removed; full archived copies remain under `docs/archives/`,
 inbound links (README, PLATFORM_ARCHITECTURE, DESKTOP_APP_PLAN) were repointed, and the sweep is
 summarized in the umbrella's **Prior audits (absorbed)**.

93 5. **HOSTING_PLAN.md** / **UI_PROPOSAL.md** are PROPOSAL but partly overtaken **by ADR-013**
 (DO dropped) + the shipped khelsutra site. **UI_PROPOSAL.md** banner added (2026-06-21) ?
 its UX half is now tracked in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; the `I18N\_PLAN.md`
 localization track is likewise in implementation there. **HOSTING_PLAN.md** banner still
 pending (ADR-013 already noted in it).

94 6. **RESOLVED** (2026-06-21) ? **COMPUTE_DECOUPLED_SERVING** M0 approved + M1?M4 SHIPPED
 (#279/#280/#282/#283); all §8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized
 (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17
 \$100/?10k cap).

95

96 **## 5. Maintenance**

97 Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at
 each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md`
 archive policy and [[working-agreement-tracked-project-docs]].

98

13. user (2026-07-03T16:40:45.315Z)

1 # Next steps (live) ? refreshed 2026-06-21

2

3 > **How to read:** this top section is the **current cross-track state + prioritized**
 pending list (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed
owned-model & rally-quality track plan (2026-06-13) is **preserved below, still valid** ? it
 is now **one of several** active tracks. Live blow-by-blow: memory `current-state.md`. Doc
 lifecycle/hygiene: `docs/DOC\_STATUS.md`.

4

5 **## ? Where we are** (2026-06-21)

6 **DECOUPLED_COMPUTE** delivered on GCP + archived (#270); **compute = GCP only**
 (ADR-013, DO dropped). **COMPUTE_DECOUPLED_SERVING** subproject spun up (#272, **ADR-014**) ?
 one pure core + local/cloud profiles; **M0?M9 LARGELY SHIPPED** (ADR-014 ratified 2026-06-21;
 M1?M4 #279?#283, M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF). **Per-video**
 workspace P0 shipped (#264); **doc-status register + archival due-diligence** in place (#273).
 The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality /**
owned-model track (below) is the core ML work, gated on **growing the golden corpus**.

7

8 > **LAUNCH DECISION ? NO-GO** (owner, 2026-06-30). **The rally-detection product is**
not cleared for an external/commercial launch; we stay in **development**. This is the
 expected default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the
ship-gate target (#172) ? currently unset, needs corpus growth (and the R2 tolF1 ablation,
 item 8); **(2)** resolve the **WASB shuttle-detector weight provenance** (commercial-clean
 licensing). P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev
 PRs ? launching. Re-decide only when both unblockers clear.

9

10 **## ? Prioritized pending ? all tracks**

11 **P0 ? highest leverage / unblocks the most**

12 0. **??** [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD\_GPU\_DRYRUN\_GUIDE.md`); **M12** does NOT fix it.** Needs (a separate khelsutra `web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a ***persistent*** endpoint), **(ii)** the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9** edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER** in the UI** ("your machine vs our cloud", D13/D15). The engine seams make it ***possible***; the SPA makes it ***usable***. Full record: **[[ui-driven-cloud-serving-gap]]**. **Concrete execution tracker** after the n=15 corpus review:** `docs/ALPHA\_LAUNCH\_READINESS.md` §7.

13 0b. **??** [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING\_PLAN.md`.** The **LIGHT-tier** engine is bundle-ready: `pip install` ? `indexer --app` serves the ***bundled SPA*** single-origin, ***torch-free***, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **Next** (fresh focused pushes): **(1)** P3 **GPU validation + screencast** (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]` ; ? native-WASB 0-rally tuned-config blocker), **(2)** P2b-install (uv/script + truly-local default config), **(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

14 1. ? **COMPUTE_DECOUPLED_SERVING** M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), **M5** #286, M6 #287, M8 #288, M9-auth #290** all merged (default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a **\$100/?10k cap** (D17).

15 2. **[owner, ongoing]** Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track ***and*** full-corpus training feed on it.

16 3. **[owner + Claude . GPU]** Full-corpus training (n?5)** ? a real `weights/<ver>/` (the `0.1.0-l4` bundle is n=2 plumbing-only).

17

18 **P1** ? near-term, mostly Claude-doable (default-OFF, green-gated)**

19 3b. **[Claude]** Five-repo audit health remediation ? **COMPLETE** (2026-07-02).** All waves (P0?P28) landed and the last residuals were closed or filed as issues; the umbrella tracker was archived to `docs/archives/past\_projects/AUDIT\_REPORT\_khelsutra-guru\_2026-07-02-COMPLETED-2026-07-02.md`. Remaining owner-gated/future work lives as GitHub issues: #443 (CORS default), #444 (corpus Phase-2 upload), #445 (GCS eval plumbing), plus #301/#327/#332/#384.

20 4. **[Claude]** **COMPUTE_DECOUPLED_SERVING** batch ? REMAINING: M11 + M12** (M5/M6/M8/M9-auth ? merged this session). **M11** = `backend/flywheel.py` capture?store?shadow-eval?goldenize plumbing reusing `workspace.py::for\_video` + `backend/eval/experiment.compare` + `backend/eval/promotion.py::promote\_if\_served\_safe`, behind `flywheel.\*`=OFF + a faked `consent\_attested` (? privacy-sensitive ? give it fresh focus). **M12** = `backend/storage/gdrive\_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme (regex already allows the hyphen; add to `\_KNOWN` + `\_backend\_class` + the StorageConfig Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://` untouched. ? **M9-consent cols** = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.

21 5. **[Claude]** Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/+`tmp/` \*[[the `output/chunks_` de-hardcode base = ? done via #282]]* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.

22 6. ? **Ops-Agent** codified (#291)** ? `deploy/gcp/create\_gpu\_vm.sh` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. **PROVEN** active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])

23 7. **[owner]** Set the ship-gate target (#172)** ? needs corpus growth.

24 8. **[owner-side GPU]** R-series: promote-or-reject `R2` tolF1 as the gate** (the ablation) ? needs GPU-extracted golden features.

25

26 **P2** ? later / gated / deferred**

27 9. Rally-quality **P3/P4** (person-fusion LOV0, audio cue) ? needs GPU golden features.

28 9b. ? **DONE** (#396):** the R4 density fix (finding B / P2b in

```

`RALLY_DETECTION_FIXES_PLAN.md`) is **promoted to default-ON**. Golden-set A/B at the **SERVED**
windowing showed tolF1 0.353?0.364 (?tolF1 +0.010, no regression), so
`inactivity_temporal_density` default was flipped to True. (Landed flag-gated default-OFF in
#394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the
rally tail open.
29     10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
30     11. **Multisport** (table-tennis first, then pickleball).
31     12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within
`COMPUTE_DECOUPLED_SERVING`; owner/legal-gated).
32     13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.
33     14. **Rename repo off "badminton"* (owner, BACKLOG).
34     15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector
**md5** export (collector repo).
35     16. **PMF field validation** (owner-side, business track).
36
37     **Doc-hygiene (tracked in `docs/DOC_STATUS.md` §4): ** ? #1 per-video reconcile + ? #2
this refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary
confirm); confirm the COMPLETED top-level stubs; verify `HOSTING_PLAN`/`UI_PROPOSAL` currency.
38
39     ---
40
41     ## Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)
42
43     **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
44     owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
45     ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
46     `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;  

**PR-2
47     upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
48     compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
49     coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
50     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
51     **Authoritative roadmap:**
52     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
53     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
54
55     ## ? Next-session / GPU-box pickup (2026-06-15)
56     **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
57
58     **Immediate next step ? the real P3 LOVO measurement** (this box has a GPU +
`torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden
features + `.rallies.csv` labels are not yet here):
59     1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `*_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
60     2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.

```

```

61      3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
62
63      **Also queued (owner-gated / pending input):**
64      - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
65      - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
66      - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
67
68      **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
69
70      ## Where we are (one paragraph)
71      ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
72      behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
73      the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
74      eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
75      sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
76      (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
77      battery is done** (see table) ? the moat is quantified.
78
79      ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
80      | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
81      |---|---|---|
82      | Heuristic (velocity windowing) | 0.657 | 0.157 |
83      | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
84      | Two-stage WASB+Gemini refine | 0.83 | ? |
85      | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
86
87      **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
88      wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
89      the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
90      architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
91      only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
92      Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
93
94      ## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation
in progress)
95      **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR
#112 + session). All implementation is **owner-gated** (per explicit in the plan); each step =
ONE PR off `master`.
96
97      **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs
#156/#157/#160/#161/#163 + #165 + final records).** See archived
`docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every
pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules
+ STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive
done; clean slate.
98

```

```
99      - **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion
layer) .
100      `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
```

14. assistant (2026-07-03T16:40:47.817Z)

Now let me check more details about the new file and the conventions it should follow, particularly around documentation standards mentioned in CLAUDE.md.

15. user (2026-07-03T16:40:49.024Z)

```
1      # Alpha Launch Readiness Plan
2
3      > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-03
4      > **Lifecycle:** `DRAFT -> IN PROGRESS -> DONE -> archived`
5      > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
6      > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`,
`COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.
7      > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
8
9      ---
10
11     ## 0. TL;DR
12
13     The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before
manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion
schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed
before it can gate anything.
14
15     The first controlled alpha should be a hosted, authenticated product loop: upload a
video, process it, select rallies, compile, and download a reel. The owned-model/commercial
rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the
WASB checkpoint provenance is resolved.
16
17     ## 1. Problem & goal
18
19     The previous launch blockers were spread across roadmap docs and mixed together three
different questions:
20
21     - **Can an invited tester use the product without SSH?**
22     - **Can the 15-video golden corpus now support honest quality gates?**
23     - **Can we claim or ship an owned/commercial rally detector?**
24
25     This doc separates those into executable workstreams. The goal is to reach an invite-
only alpha where the product is usable end-to-end, the quality claims are honest and
reproducible, and the remaining model/legal gates are explicit rather than accidentally implied
by launch activity.
26
27     ## 2. Decisions locked
28
29     | # | Decision | Source / date | Implication |
30     |---|---|---|---|
31     | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled
hosted product loop without claiming the owned model is commercially cleared. |
32     | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
```

```

33 | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature
completeness, baseline refresh, and gate ratification come before quality claims. |
34 | D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-
windowing safety; keep it as an opt-in/analysis lever. |
35 | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens
D13.** | sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha
can use manual/offline corpus promotion; automated flywheel follows after serving and gates are
stable. |
36
37 ## 3. Verified snapshot, 2026-07-03
38
39 All five Git checkouts were fast-forwarded before analysis:
40
41 | Repo | Branch | Synced HEAD |
42 |---|---|---|
43 | `badminton-highlight-indexer` | `master` | `e4914d1` |
44 | `khelsutra` | `master` | `b0f70a5` |
45 | `rally-annotator` | `main` | `ef9e088` |
46 | `sports-data-collector` | `master` | `bfc014f` |
47 | `sports-obsreport` | `main` | `a0c7252` |
48
49 Corpus/eval facts:
50
51 - `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says
`output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
52 - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
found all 15 trajectories and labels in this workspace.
53 - `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
54 - `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only
6 videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
55 - `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
56 - `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
57 - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
58 - `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity
group` currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
59
60 ## 4. Launch strategy

```

16. user (2026-07-03T16:40:49.772Z)

```

cd07339 docs: add alpha launch readiness tracker
e4914d1 Surface validation failure reasons
f0814d5 Normalize upload filenames before extension checks
e585965 Fix ffmpeg concat clip paths (#449)
ac6184b docs: archive the completed 2026-07-02 workspace audit umbrella (#446)

```

17. assistant (2026-07-03T16:40:52.002Z)

Now let me check the CLAUDE.md file more carefully regarding documentation standards and any memory/linking conventions.

18. user (2026-07-03T16:40:52.715Z)

```

70 ## Workflow conventions

```

```

71 - **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
72 - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.10+ (the `trackers`
package requires ?3.10); `ffmpeg` required.
73 - Never push to the default branch directly; descriptive commit messages.
74 - **Substantial design/project work gets a tracked project doc** following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` §7,
`docs/GOLDEN_REGRESSION_FIXTURES.md`.
75 - **Launch / "bit-flip" PRs** (a default-flip that drives a tracked project to terminal
state) MUST, in the SAME PR, run the `docs/DOC_STATUS.md` §2 archival due-diligence on that
project's docs (verify-vs-code ? flip status + completion note ? update inbound refs ? keep ADR
lineage ? `git mv` to `docs/archives/` ? record in DOC_STATUS §3), and fix any internal-link
breakage the move causes. Enforced by `.github/PULL_REQUEST_TEMPLATE.md` + the `link-check` CI
job (`python -m tools.check_md_links`).
76 - ?? Hand the heaviest machine-side checks (real video, the full GPU/WSL/WASB suite) to
the owner,
77     who has the golden corpus + labelled videos; verify pure-logic paths locally.

```

19. user (2026-07-03T16:40:54.939Z)

(Bash completed with no output)

20. assistant (2026-07-03T16:40:57.756Z)

Now let me check the project doc template to see what conventions should be followed.

21. user (2026-07-03T16:40:57.984Z)

```

1      <!--
2      PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3
4      A "tracked project doc" is how this repo carries substantial design/project work: it can
be
5      ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables,
then
6      archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
7      docs/GOLDEN_REGRESSION_FIXTURES.md.
8
9      Rules of thumb:
10     - §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable
PR.
11     - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs
sign-off),
12       or [design] (proposed). The doc reflects real shipped state, never aspirational.
13     - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
14     - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model
for
15       research- or security/privacy-heavy work). Delete sections that do not apply.
16     - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state
work).
17     -->
18
19     # <Project / Plan Title>
20
21     > **Status:** `DRAFT` ? `<owner-gated? | in progress>` . **Owner:** `<name>` .
**Created:** `<YYYY-MM-DD>` . **Last updated:** `<YYYY-MM-DD>`
22     > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
23     > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
24     > **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
25     > **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.

```



```

26
27 ---
28
29 ## 0. TL;DR
30 <2?5 sentences: the goal, the approach, and the single most important caveat. Call out
any finding that is already actionable on its own.>
31
32 ## 1. Problem & goal
33 <Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a
reader must read to get current.>
34
35 ## 2. Decisions locked
36 <Table ? capture answered questions so they are not relitigated.>
37
38 | # | Decision | Source / date | Implication |
39 |---|-----|-----|-----|
40 | D1 | ? | ? | ? |
41
42 ## 3. Foundation ? research / prior art *(OPTIONAL)*
43 <Include when the plan rests on external facts: cited research, legal/compliance,
benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this
creates.>
44
45 ## 4. Design / architecture
46 <The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing
modules/code this builds on so it is clear what is new vs reused.>
47
48 ## 5. Threat model / risk table *(OPTIONAL ? security / privacy / compliance work)*
49 <Who/what is protected vs NOT; residual risks stated plainly.>
50
51 ## 6. Honest limits ? what this does NOT do
52 <The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>
53
54 ## 7. Deliverables & progress tracker ? **source of truth**
55 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
56
57 | ID | Deliverable | Depends on | Gated? | Status | PR |
58 |---|-----|-----|-----|-----|----|
59 | P0 | ? | ? | No | ? | ? |
60
61 ## 8. Open questions ? owner / external *(split blocking-gates from nice-to-knows; name
who decides)*
62 1. ?
63
64 ## 9. Definition of done
65 <Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>
66 - [ ] ?
67
68 ## 10. References
69 **Internal code anchors `[verified]`:** `<paths>`. **Internal docs:** `<docs>`.
**External:** `<cited sources>`.
70
71 ---
72
73 ### Changelog
74 - `<YYYY-MM-DD>` ? `<what changed>`.
75

```

22. user (2026-07-03T16:40:59.170Z)

[check-md-links] clean ? no broken internal links in live docs.

23. assistant (2026-07-03T16:41:01.439Z)

Excellent! The link check passed. Now let me review the changes more carefully to look for violations. Let me examine each file change in the diff and check specific lines against the conventions in CLAUDE.md.

24. user (2026-07-03T16:41:02.895Z)

```
diff --git a/docs/README.md b/docs/README.md
index 0df1a69..58be5ff 100644
--- a/docs/README.md
+++ b/docs/README.md
@@ -15,6 +15,7 @@
- [DOC_STATUS.md](DOC_STATUS.md) ? **? living Documentation Status & Archival Register**: every
foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence
checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here;
update rows as docs complete.

## Product / platform plans
+- [ALPHA_LAUNCH_READINESS.md](ALPHA_LAUNCH_READINESS.md) ? **? Active tracked project (`DRAFT`,
created 2026-07-03):** cross-repo alpha readiness tracker after the golden corpus reached 15
videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel) from the
owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved). S7 is the
source of truth for the concrete next PR sequence: corpus portability, n=15 baseline refresh,
ablation/fusion reruns, hosted serving, alpha copy/launch review.
- [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked
project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to
DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never
branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend`
(what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU,
in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run
GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI).
Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY
proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a
[runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage
ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). S7 M0?M10 tracker is the source
of truth.
- [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED +
ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing
bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB
runner**, **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ?
DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving
endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving"
subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under
`docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).
- [PACKAGING_PLAN.md](PACKAGING_PLAN.md) ? **? Active tracked project (`IN PROGRESS`, 11 PRs
merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ?
`pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA**
single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision
D10/01). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP_APP_PLAN.md`
narrative.** S7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **?
Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4
(ONNX/train/annotate = north star). Memory `[packaging-app-plan]` / `[next-session-packaging-
app]`.
```

25. user (2026-07-03T16:41:03.681Z)

```
diff --git a/docs/DOC_STATUS.md b/docs/DOC_STATUS.md
index 34411fd..2541672 100644
--- a/docs/DOC_STATUS.md
+++ b/docs/DOC_STATUS.md
@@ -30,6 +30,7 @@
@@ -30,6 +30,7 @@ Before a foundational/tracked doc moves to `docs/archives/`:
### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's S7 is the completeness
source-of-truth)
| Doc | Status | Archive-ready? | Next action |
| ---|---|---|---|
+| `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created
```

2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-model/commercial rally-detection gates | No | Execute S7 in order: corpus portability + n=15 baselines, hosted alpha serving, ablation/fusion reruns, then launch GO/NO-GO |
`COMPUTE\_DECOUPLED\_SERVING/`	**IN PROGRESS** ? M0 approved + **M1?M9 LARGELY SHIPPED** (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights pushed (`weights/0.1.0-l4/`)	No	M11 flywheel + M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap)
`RALLY\_QUALITY\_RESEARCH.md`	Foundation, **tracked & ACTIVE** (S7)	No	Drives the R-series; complete when S7.1 single quality number is attributed
`R2\_EVAL\_METRIC\_DESIGN.md`	IMPLEMENTED (augmenting)	No (gates R4 + dual-eval; ablation pending)	Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster

26. user (2026-07-03T16:41:04.405Z)

```
diff --git a/docs/NEXT_STEPS.md b/docs/NEXT_STEPS.md
index f186ff9..a9f35a8 100644
--- a/docs/NEXT_STEPS.md
+++ b/docs/NEXT_STEPS.md
@@ -9,7 +9,7 @@
```

```
## ? Prioritized pending ? all tracks
**P0 ? highest leverage / unblocks the most**

-0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only**
(`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/`
SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service**
(M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA
ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll
`/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-
OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our
cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:
[[ui-driven-cloud-serving-gap]].

+0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only**
(`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/`
SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service**
(M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA
ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll
`/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-
OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our
cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:
[[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:**
`docs/ALPHA_LAUNCH_READINESS.md` S7.

0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-
included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install`
? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state
/ ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b,
P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** ** (1)** P3 **GPU validation +
screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB
0-rally tuned-config blocker), ** (2)** P2b-install (uv/script + truly-local default config),
** (3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud
(Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-
session-packaging-app]]`.

1. ? **`COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified).
M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-
OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All S8 gating Qs
locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA
(M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF
team-domain/AUD + ingress-lock; real cloud verification authorized within a **$100/?10k cap**
(D17).

2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT
first). The single highest-leverage move; the quality track *and* full-corpus training feed on
it.
```

27. assistant (2026-07-03T16:41:07.748Z)

Perfect! Now I have the full diff. Let me check specifically for any violations against the documented conventions. Let me look at a few key areas:

1. Check if the new document follows the PROJECT_DOC_TEMPLATE conventions properly
2. Verify the honesty tags (`[verified]` vs `[analysis]`)
3. Check for memory links convention (`[...]`)

28. user (2026-07-03T16:41:08.761Z)

```
1      # Alpha Launch Readiness Plan
2
3      > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-03
4      > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`
5      > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
6      > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`,
`COMPUTE_DECOUPLED_SERVING/`, and the sibling `khehsutra` Studio docs.
7      > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
8
9      ---
10
11     ## 0. TL;DR
12
13     The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before
manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion
schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed
before it can gate anything.
14
15     The first controlled alpha should be a hosted, authenticated product loop: upload a
video, process it, select rallies, compile, and download a reel. The owned-model/commercial
rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the
WASB checkpoint provenance is resolved.
16
17     ## 1. Problem & goal
18
19     The previous launch blockers were spread across roadmap docs and mixed together three
different questions:
20
21     - **Can an invited tester use the product without SSH?**
22     - **Can the 15-video golden corpus now support honest quality gates?**
23     - **Can we claim or ship an owned/commercial rally detector?**
24
25     This doc separates those into executable workstreams. The goal is to reach an invite-
only alpha where the product is usable end-to-end, the quality claims are honest and
reproducible, and the remaining model/legal gates are explicit rather than accidentally implied
by launch activity.
26
27     ## 2. Decisions locked
28
29     | # | Decision | Source / date | Implication |
30     |---|---|---|---|
31     | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled
hosted product loop without claiming the owned model is commercially cleared. |
32     | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
33     | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature
```

```

completeness, baseline refresh, and gate ratification come before quality claims. |
34 | D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-
windowing safety; keep it as an opt-in/analysis lever. |
35 | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens
D13.** | sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha
can use manual/offline corpus promotion; automated flywheel follows after serving and gates are
stable. |
36
37 ## 3. Verified snapshot, 2026-07-03
38
39 All five Git checkouts were fast-forwarded before analysis:
40
41 | Repo | Branch | Synced HEAD |
42 |---|---|---|
43 | `badminton-highlight-indexer` | `master` | `e4914d1` |
44 | `khelsutra` | `master` | `b0f70a5` |
45 | `rally-annotator` | `main` | `ef9e088` |
46 | `sports-data-collector` | `master` | `bfc014f` |
47 | `sports-obsreport` | `main` | `a0c7252` |
48
49 Corpus/eval facts:
50
51 - `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says
`output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
52 - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
found all 15 trajectories and labels in this workspace.
53 - `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
54 - `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only
6 videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
55 - `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
56 - `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
57 - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
58 - `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity
group` currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
59
60 ## 4. Launch strategy
61
62 ### 4.1 Alpha lane: make the product usable
63
64 The alpha lane is successful when an invited tester can open the app, authenticate,
upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
should avoid model-quality promises beyond the measured state and should default to the most
reliable serving path.
65
66 ### 4.2 Quality lane: make the 15-video corpus gate-ready
67
68 The quality lane is successful when the 15-video corpus is portable, reproducible,
feature-complete, and backed by refreshed baselines. Only then should R2 tolF1,
Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.
69
70 ### 4.3 Flywheel lane: make tester feedback useful
71
72 The flywheel lane is successful when user corrections can flow into the collector/vault
path with consent/provenance and a visible re-eval delta. This is valuable for alpha learning,
but it should not block the first controlled alpha if manual promotion can bridge the gap.
73

```

```

74     ## 5. Risks and guardrails
75
76     | Risk | Why it matters | Guardrail |
77     |---|---|---|
78     | Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
79     | Stale local manifests create false confidence | Manifest-driven tools skipped or
failed until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke
check |
80     | 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still
n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
81     | Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly
reported stale baselines | Refresh baselines only after reviewer sign-off and record the
command/output |
82     | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but
can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha
lane is already stable |
83     | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained
model regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product
launch tasks |
84
85     ## 6. Honest limits
86
87     Completing this doc's alpha lane does **not** prove the owned model is ready, that WASB
weights are commercially clean, that the product can serve unbounded public traffic, that multi-
sport is ready, or that every tester correction automatically improves the model.
88
89     Completing the quality lane does **not** by itself deploy a service. It only makes the
measurement surface honest enough for a launch decision.
90
91     ## 7. Deliverables & progress tracker
92
93     Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the
row explicitly names a cross-repo batch.
94
95     | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
96     |---|-----|-----|-----|-----|-----|---|
97     | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-
indexer` | - | No | IN PROGRESS | #451 |
98     | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check,
edge auth ON, upload -> process -> jobs -> compile -> download verified | `khealsutra` +
`badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
99     | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke
check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer`
| A0 | No | TODO | - |
100    | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
collector` + vault | A2 | Owner upload/storage scope | TODO | - |
101    | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | TODO | - |
102    | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6`
floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |
103    | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation
on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
104    | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | TODO | - |
105    | A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
106    | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce
a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner

```

```

approval for M0.4 scope | TODO | - |
107 | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum
threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-
indexer` | A4, A5, A8 | Owner decision | GATED | - |
108 | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
109 | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
promotion, `train|reeval` job, and delta-F1 surface | `khelesutra` + `badminton-highlight-
indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |
110 | A13 | Alpha launch review: runbook, privacy/consent copy, known limits,
rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch
decision | TODO | - |
111
112 ## 8. Concrete next PR sequence
113
114 1. **Docs PR:** land this tracker.
115 2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke
test.
116 3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download
verification and runbook updates.
117 4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor
refresh.
118 5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
119 6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy
limits, and launch checklist.
120
121 ## 9. Open questions
122
123 1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or
should P10/P11 be reopened before any tester wave?
124 2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU,
GCE L4 VM, or another approved GPU host?
125 3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains
unresolved?
126 4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
127 5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every
clip, or should the eval loaders support mixed schemas with explicit capability tags?
128
129 ## 10. Definition of done
130
131 - [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without
SSH.
132 - [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
133 - [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
134 - [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
135 - [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
136 - [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
on the 15-video corpus.
137 - [ ] Ship-gate target and allowed alpha claims are recorded.
138 - [ ] WASB C3 status is recorded and reflected in alpha messaging.
139 - [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for
alpha-plus.
140 - [ ] Owner records GO/NO-GO for the first alpha wave.
141
142 ## 11. References
143
144 Internal docs:
145
146 - `docs/NEXT_STEPS.md`
147 - `docs/ROADMAP.md`

```

```

148 - `docs/data_pipeline/GOLDEN_VIDEOS.md`
149 - `docs/R2_EVAL_METRIC_DESIGN.md`
150 - `docs/RALLY_DETECTION_FIXES_PLAN.md`
151 - `docs/COMPUTE_DECOUPLED_SERVING/README.md`
152 - sibling `khelsutra/NEXT_STEPS.md`
153 - sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
154 - sibling `khelsutra/docs/PER_RALLY_SELECTION.md`
155 - sibling `sports-data-collector/data/golden_source_videos.json`
156
157 Local verification commands used for the 2026-07-03 snapshot:
158
159 ```powershell
160 python -m backend.eval.fusion_compare --features-dir output
161 python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
162 python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor
eval_baselines\heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
163 python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir
output --no-report
164 python -m backend.eval.ablation --features-dir output --granularity group
165 ```
166
167 ---
168
169 ### Changelog
170
171 - 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`,
golden-corpus docs, local eval artifacts, and the current 15-video corpus state.

```

29. assistant (2026-07-03T16:41:11.297Z)

Now let me check if there are any memory link conventions that should have been used but weren't. Let me also verify if the section numbering matches the template.

30. user (2026-07-03T16:41:12.414Z)

(Bash completed with no output)

31. assistant (2026-07-03T16:41:14.870Z)

Good - no memory links are present. Now let me check if the NEXT_STEPS.md addition should have used a memory link. Let me check how memory links are used in that file:

32. user (2026-07-03T16:41:15.689Z)

```

12:0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only**
(`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/`
SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service**
(M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA
ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll
`/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-
OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our
cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:
[[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:**
`docs/ALPHA_LAUNCH_READINESS.md` §7.
13:0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-
included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install`
? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state
/ ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b,
P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** ** (1) ** P3 **GPU validation +
screenrecast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB
0-rally tuned-config blocker), ** (2) ** P2b-install (uv/script + truly-local default config),
** (3) ** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud
(Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-

```



```
session-packaging-app]]`.
22:6. ? **Ops-Agent codified (#291)** ? `deploy/gcp/create_gpu_vm.sh` (the one-command VM
helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN
active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])
```

33. assistant (2026-07-03T16:41:18.148Z)

Good - memory links are properly used in NEXT_STEPS.md. Let me now check line 12 more carefully to see the exact change made in the diff:

34. user (2026-07-03T16:41:18.635Z)

```
9
10     ## ? Prioritized pending ? all tracks
11     **P0 ? highest leverage / unblocks the most**
12     0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few
recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-
only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra
`web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable
**service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)**
the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ?
poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290,
default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs
our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full
record: [[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus
review:** `docs/ALPHA_LAUNCH_READINESS.md` §7.
13     0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable
batteries-included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip
install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-
home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades
(P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** ***(1)** P3 **GPU
validation + screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ?
native-WASB 0-rally tuned-config blocker), ***(2)** P2b-install (uv/script + truly-local default
config), ***(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then
P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` /
`[[next-session-packaging-app]]`.
```

35. assistant (2026-07-03T16:41:32.050Z)

Perfect. Now let me verify that the sections in the ALPHA_LAUNCH_READINESS.md follow the proper template structure. Let me review the PROJECT_DOC_TEMPLATE again and compare:

The template has:

- 0. TL;DR
- 1. Problem & goal
- 2. Decisions locked
- 3. Foundation (OPTIONAL)
- 4. Design / architecture
- 5. Threat model (OPTIONAL)
- 6. Honest limits
- 7. Deliverables & progress tracker
- 8. Open questions
- 9. Definition of done
- 10. References
- Changelog

The ALPHA_LAUNCH_READINESS.md has:

- 0. TL;DR ?
- 1. Problem & goal ?
- 2. Decisions locked ?
- 3. Verified snapshot (different from template's "Foundation")
- 4. Launch strategy (different from template's "Design")
- 5. Risks and guardrails (renamed from "Threat model")
- 6. Honest limits ?

- 7. Deliverables & progress tracker ?
- 8. Concrete next PR sequence (added)
- 9. Open questions ?
- 10. Definition of done ?
- 11. References ?
- Changelog ?

This looks like a reasonable adaptation of the template. The custom section names (§3, §4, §5, §8) are justified. Let me verify the status header format matches the template requirements:

Template says:

```

...
> **Status:** `DRAFT` ? `<owner-gated? | in progress>` . **Owner:** `<name>` . **Created:**
`<YYYY-MM-DD>` . **Last updated:** `<YYYY-MM-DD>`
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`;
pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
> **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
> **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.
...

```

The file has (line 3-7):

```

...
> **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-03
> **Lifecycle:** `DRAFT -> IN PROGRESS -> DONE -> archived`
> **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed in
`docs/README.md` and `docs/DOC_STATUS.md`.
> **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`,
`COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.
> **Honesty note:** `[verified]` means checked against the local synced repos on 2026-07-03.
`[analysis]` means derived from a local command run on that snapshot. This is not legal,
financial, or launch approval.
...

```

Now let me check for the specific violations. Let me search through the CLAUDE.md rules again very carefully:

1. **Project doc rule** (line 74): "Substantial design/project work gets a tracked project doc following `docs/PROJECT\_DOC\_TEMPLATE.md` (status header · decisions-locked · progress tracker · definition-of-done; archived on completion)."

The ALPHA_LAUNCH_READINESS.md follows the template structure well.

2. **Memory linking** - I need to check if there should be a memory link to this doc. Let me check how other tracked project docs are referenced in memory:

36. user (2026-07-03T16:41:33.422Z)

```

./C--Users-avidu-Projects--NET/memory.pre-sync.bak/hellodotnet-test-coverage.md
./C--Users-avidu-Projects--NET/memory.pre-sync.bak/MEMORY.md
./C--Users-avidu-Projects-badminton-highlight-indexer/memory.pre-sync.bak/audio-e1-findings.md
./C--Users-avidu-Projects-badminton-highlight-indexer/memory.pre-sync.bak/cache-hygiene-
policy.md
./C--Users-avidu-Projects-badminton-highlight-indexer/memory.pre-sync.bak/candidate-segments-
finetuning.md

```

37. assistant (2026-07-03T16:41:35.288Z)

Let me check the memory for the badminton-highlight-indexer project to see if there should be a memory link to the new doc:

38. user (2026-07-03T16:41:36.437Z)

(Bash completed with no output)

39. user (2026-07-03T16:41:40.100Z)

```
C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer
C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer--claude-
worktrees-ablation-schema-fix
C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer--claude-
worktrees-m11-flywheel
C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer--claude-
worktrees-nightly-e2e
C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer--claude-
worktrees-pr-stitch
C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer--claude-
worktrees-refactor-base
C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer--claude-
worktrees-scripts-unwrap
C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer--claude-
worktrees-w2-safeguard
C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-khelsutra-guru-badminton-highlight-
indexer
C:/Users/avidu/.claude/projects/c--Users-avidu-Projects-khelsutra-guru-badminton-highlight-
indexer-badminton-highlight-indexer
C:/Users/avidu/.claude/projects/c--Users-avidu-Projects-Verifiers-khelsutra-badminton-highlight-
indexer-badminton-highlight-indexer
```

40. assistant (2026-07-03T16:41:42.468Z)

Let me check a simpler approach - looking at the specific CLAUDE.md rules more carefully. Now let me analyze what the actual violations might be. Let me re-read the global CLAUDE.md file more carefully for doc conventions:

41. user (2026-07-03T16:41:42.743Z)

```
1      # Global instructions (all projects)
2
3      ## Repo-freshness convention: git pull BEFORE reading
4      **Always `git pull` a repo before starting to read it** ? at session start for the
primary
5      working directory, and again for any sibling/local repo the moment work touches it.
6
7      - **Why:** the owner closes every session on a "clean slate", but multiple parallel
slates
8      (sessions/machines) exist ? PRs get merged and master moves between sessions, so the
local
9      checkout can silently lag the remote truth. Pull first so ramp-up reads (handoff,
docs, code)
10     reflect where the remote actually is.
11     - **How (safe form):** `git pull --ff-only` on the current branch (a plain fetch+ff).
Never
12     auto-merge, rebase, or discard local state to make a pull succeed.
13     - **If it can't fast-forward** (diverged branch, dirty tree in the way): do NOT force
anything ?
14     report the state to the owner and proceed read-only on what's local, flagging that it
may be
15     stale. Uncommitted changes are often a sibling session's or the owner's WIP; never
clobber them.
16
17     ## Git branching safety: concurrent sessions / shared checkouts
18     Multiple sessions ? and spawned/background tasks ? may share ONE working checkout and
create
19     branches + commit **concurrently**, so `git branch` / `git log` can change UNDER you
```

```

between two
20  commands. (Spawned "fresh worktree" tasks are NOT always isolated.) This has caused a
real
21  incident: a sibling commit landed in the gap between a branch-point check and `git
checkout -b`,
22  so the new branch rode on top of it and a **squash-merge silently absorbed the sibling's
work**.
23  Protocol ? do this EVERY time, not only when you suspect a collision:
24  - **Branch from the REMOTE, explicitly:** `git fetch origin` then `git checkout -b
<name> origin/<base>`
25  (e.g. `origin/master`). Never assume the current branch is the intended base.
26  - **Re-verify right before `git add`/commit:** `git branch --show-current` + `git log
--oneline -1`
27  (HEAD is yours). Before opening a PR, `git log --oneline origin/<base>..HEAD` must
list ONLY your
28  commits ? if a sibling commit appears, re-cut the branch from `origin/<base>`.
29  - **Stage explicit paths** (`git add <files>`) ? never `git add -A`/`.`/`-u`, so sibling
or untracked
30  files can't ride into your commit.
31  - **Serialize repo writes:** don't start a repo-writing spawned/background task while
you hold
32  uncommitted work ? commit or push first. If one may be running, treat the tree +
current branch
33  as able to shift, and put this branching-safety reminder into the spawned task's own
prompt.
34
35  ## Session-handoff convention
36  Maintain a living **session handoff** for each project and use it to resume context
quickly
37  across windows.
38
39  - **Where it lives:**
40  - If the project has an auto-memory dir (`~/.claude/projects/<project>/memory/`), keep
the
41  handoff as `session-handoff.md` there, and list it under a ***"? Start here*** entry
at the
42  top of that project's `MEMORY.md`.
43  - Otherwise, keep a `SESSION_HANDOFF.md` at the repo root.
44  - **What it contains** (keep it to ~1 page ? it POINTS to detailed artifacts, never
duplicates them):
45  1. **You are here** ? current state.
46  2. **Next steps / open threads.**
47  3. **Ramp-up kit** ? the exact files/docs to read to get current.
48  4. **Key decisions** ? so they aren't relitigated.
49  - **At session start:** if a handoff exists, read it before starting substantive work,
and use it
50  (plus its ramp-up kit) to get up to speed instead of replaying past conversation.
51  - **Updating it:** keep it current **automatically** ? refresh at meaningful checkpoints
(a PR is
52  pushed/merged, a key decision is made, work shifts phase, or a session is wrapping
up), so a
53  restart can ramp up without losing context. Do **not** rewrite it every turn ? only
when
54  something worth resuming from changed. The phrases ***"update the handoff"*** /
***"refresh the
55  handoff"*** force an immediate full rewrite on demand.
56  - **Keep it honest:** it reflects real, shipped state ? never aspirational. (See each
project's
57  attribution/working-agreement note where present.)
58
59  ## Code review ? pre-LGTM gate (applies to every PR/code review, every project)
60
61  Do **not** post **LGTM** or otherwise sign off on a pull request until I have personally
**verified** the following ? never trust the PR description's claims, reproduce them locally:
62

```

```

63     1. **Full test suite is green.** Run the whole suite (e.g. `run_all_tests.py` /
`pytest`), not just the changed files.
64     2. **Install any dependency needed** so the suite actually collects and runs ? do not
let modules skip or error for missing deps. (If a check genuinely can't run locally ?
GPU/WSL/real-video/etc. ? say so explicitly and note what was and wasn't verified.)
65     3. **Lint + type checks are clean** ? `ruff check .` and `mypy ...`, matching the repo's
CI. Install `ruff`/`mypy` if absent.
66     4. **Coverage stays up** ? run with coverage and confirm it stays **at or above the
repo's floor** (e.g. the CI `--cov-fail-under` value) and does **not regress** versus the base
branch.
67     5. **Review findings are resolved** ? bugs and repo-convention/discipline issues
addressed.
68
69     LGTM is a *verified* stamp, never a rubber stamp. State plainly what was and wasn't
verifiable locally.
70

```

42. assistant (2026-07-03T16:41:46.669Z)

Now let me carefully review the PR to check for violations against the rules I've found. Let me look at a few specific areas:

1. The new file ALPHA_LAUNCH_READINESS.md section 7 has "PR" column with "#451" - let me verify this matches the current branch state
2. Section 3 is labeled "Verified snapshot, 2026-07-03" but the template calls this "Foundation"
3. Check the honesty tags carefully

43. user (2026-07-03T16:41:48.050Z)

```

codex/alpha-readiness-plan
cd07339 docs: add alpha launch readiness tracker

```

44. assistant (2026-07-03T16:42:02.149Z)

Good. Now let me see if the ALPHA_LAUNCH_READINESS file claims this is "IN PROGRESS" when it should match the actual status:

The file says (line 97 in section 7):

```

...
| A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-indexer`
| - | No | IN PROGRESS | #451 |
...

```

But the status header (line 3) says:

```

...
> **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-03
...

```

This is potentially a violation! The template says section 7 uses legend "TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/verified, GATED = blocked on owner/legal/platform decision."

If this PR #451 is just landing the doc (creating it), should A0's status be "DONE" instead of "IN PROGRESS"?

Let me check the PROJECT_DOC_TEMPLATE again to understand the intended semantics:

Template line 55: "Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. ****One small PR per row****, branched from `origin/master`, no stacking."

This says "One small PR per row" - so A0 should probably be marked DONE since PR #451 lands this tracker. Let me verify what "shipped/verified" means in this context by looking at other tracked project docs:

45. user (2026-07-03T16:42:03.456Z)

7. Deliverables & progress tracker ? **source of truth**

Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One **small PR per row**, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.

46. user (2026-07-03T16:42:10.237Z)

7. Deliverables & progress tracker ? **source of truth**

Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.

ID	Deliverable	Depends on	Gated?	Status	PR
P0a	Fix `_spread` zero-collapse (behaviour-neutral, unconditional)			?	No ? #394
P0b	`_trim_inactive_tail` temporal density behind `inactivity_temporal_density` flag (default-OFF)			?	No ? #394
P1	Warn on rows silently skipped by `load_canonical_csv` (sibling `sports-data-collector`)			?	No ? sdc#50
P2a	Expose the flag in the `rally_seg_eval` A/B CLI (`--[vs-]inactivity-temporal-density`) so the promotion run is one command			P0b	No ? this PR
P2b	**Promote P0b**: golden-set A/B at SERVED windowing ? flip flag default-ON (no regression)			P2a	owner? ? #396

8. Open questions ? owner / external

- ~~Do we want `\_trim\_inactive\_tail` to fall back to the existing behavior if there are very few frames, or strictly enforce a minimum temporal density?~~ **Resolved (#393):** strict temporal density. Because `group` carries only active frames, the temporal denominator equals the sample-count denominator under dense tracking, so this is behaviour-preserving for well-tracked

47. assistant (2026-07-03T16:42:22.606Z)

I see! In RALLY_DETECTION_FIXES_PLAN.md, "P2a" says "Expose the flag in the `rally\_seg\_eval` A/B CLI... so the promotion run is one command | P2a | No | ? | this PR"

It uses "this PR" for the row that LANDS the doc itself. And the status is ? (DONE).

So in ALPHA_LAUNCH_READINESS.md, the row A0 says "Create this alpha readiness tracker and index it in docs" with status "IN PROGRESS" and PR "#451". But if PR #451 is the CURRENT PR that lands this doc, then A0 should be marked ? DONE, not IN PROGRESS.

Let me check the exact diff again to be sure:

The diff shows that line 97 in the new file states:

A0	Create this alpha readiness tracker and index it in docs	`badminton-highlight-indexer`
-	No	IN PROGRESS #451

This looks like a potential violation. The rule from CLAUDE.md (line 74) says: "Substantial design/project work gets a tracked project doc following `docs/PROJECT\_DOC\_TEMPLATE.md`" and the template (line 55) says "One small PR per row" and legend uses ?/?/?/?.

However, looking at the template more carefully (line 55-56): "Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands."

This says "Update the row (status + PR link) **as work lands**". So if PR #451 is landing this row's work (creating and indexing the tracker), the status should be ? DONE.

Let me also check the legend in the ALPHA_LAUNCH_READINESS.md file itself (line 93):

```

Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the row explicitly names a cross-repo batch.

```

It uses different legend names (TODO/IN PROGRESS/DONE/GATED) instead of the template symbols (??/??/?). This is interesting but not necessarily a violation - the document customizes the legend.

However, if A0's work (create and index the tracker) is being shipped in PR #451, then A0's status should be "DONE", not "IN PROGRESS".

This appears to be the violation. Let me make sure by finding a clear rule statement:

From CLAUDE.md (project doc template, line 55-56):

"Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. ****One small PR per row****, branched from `origin/master`, no stacking. Update the row (status + PR link) ****as work lands****."

The phrase "as work lands" means when the PR is merged/pushed. Since PR #451 is landing A0's work, A0 should be marked DONE.

Now let me check for any other violations. Let me look at the honesty tags more carefully. The template says (line 25): "mark each claim `[verified]` / `[researched]` / `[design]`"

The ALPHA_LAUNCH_READINESS file has a custom honesty note (line 7):

```

> **\*\*Honesty note:\*\*** `[verified]` means checked against the local synced repos on 2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not legal, financial, or launch approval.

```

So the document uses `[verified]` and `[analysis]` instead of `[verified]` / `[researched]` / `[design]`. Let me check if this violates the template requirement. The template says "mark each claim" - but does it have to use those exact tags?

Actually, the honesty note explicitly defines what `[verified]` and `[analysis]` mean for THIS document. That seems like a reasonable adaptation. The template example shows these are customizable per doc.

Let me now do a comprehensive check for any clear violations. Let me make one final check - are there any internal link issues or broken memory links?

48. user (2026-07-03T16:42:23.899Z)

[check-md-links] clean ? no broken internal links in live docs.

49. assistant (2026-07-03T16:42:34.954Z)

Excellent, no broken links. Now let me verify my finding about A0's status. Let me look at what it means for a row to have a PR number:

Looking at RALLY_DETECTION_FIXES_PLAN.md again, I see rows with ? DONE status have PR numbers like #394, #396, "sdc#50", "this PR".

When the template says "Update the row (status + PR link) as work lands", it means:

- When work ships in a PR, mark the row with ? DONE and add the PR number
- If the doc itself is being landed in the SAME PR, use "this PR" or the actual PR number

So A0 in ALPHA_LAUNCH_READINESS.md which says "Create this alpha readiness tracker and index it in docs" with status "IN PROGRESS" and PR "#451" appears to have an inconsistency:

Either:

1. The status should be "DONE" (because #451 is landing it), or

2. The PR should be "-" and status should be "IN PROGRESS" (because this doc PR hasn't landed yet)

Since we're reviewing PR #451 (the PR that lands this doc), and the row references #451, the row should be DONE.

This is my main finding. Let me verify there are no other violations by checking the exact wording of the CLAUDE.md rule one more time:

From CLAUDE.md line 74-75: "**Substantial design/project work gets a tracked project doc** following `docs/PROJECT\_DOC\_TEMPLATE.md` (status header · decisions-locked · progress tracker · definition-of-done; archived on completion)."

This says the doc should follow the template. The template's §7 (Progress Tracker) legend says (line 55): "Update the row (status + PR link) **as work lands**."

So yes, if A0's work (creating and indexing the tracker) lands in #451, then A0 should be marked DONE in §7.

```
```json
[
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": 97,
 "summary": "Progress tracker row A0 should be DONE, not IN PROGRESS, since PR #451 ships this work",
 "failure_scenario": "The row states status='IN PROGRESS' with PR='#451', but per PROJECT_DOC_TEMPLATE.md §7 legend 'Update the row (status + PR link) as work lands', a shipped PR should have status=DONE. Either mark A0 as DONE (if #451 ships it) or change PR to '-' and status to TODO (if unstated)."
```

```
]
```

[]
```